

David R. Rice

CONTACT INFORMATION

Department of Astronomy
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Madison, WI, USA

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RESEARCH INTERESTS

Planet Interiors, Dynamics, and Formation; Exoplanet Demographics; Computational Techniques; Geology & Material Physics; Interdisciplinary Collaborations

APPOINTMENTS

Distinguished Research Fellow Jan. 2026–Present
University of Wisconsin-Madison, Department of Astronomy
Rise-AI Initiative

Postdoctoral Researcher Aug. 2023–Jan. 2026
Astrophysics Research Center of the Open University of Israel (ARCO)
Research Group: Allona Vazan

EDUCATION

University of Nevada, Las Vegas, Las Vegas, NV
Ph.D. in Astronomy May 2023
Dissertation Advisor: Jason H. Steffen
Dissertation Title: *Inferring the compositions and interior structures of small planets*
M.S. in Astronomy Dec. 2019
M.S. Thesis Title: *The timing of dynamical interactions between planets*
Northwestern University, Evanston, IL
B.A. in Integrated Science with Honors, Physics & Astronomy, and Earth & Planetary Sciences (triple major) Jun. 2016
Honors Thesis Advisor: Fred Rasio
Honors Thesis: *Revisiting the relationship between orbital spacing and the time until an instability in planetary systems*

MENTORING

Krishang Mittal, Undergraduate Aug. 2025–Present
Srivishal Sudharsan, High School Dec. 2022–Sep. 2023
Tristan Benally, Undergraduate Aug. 2021–Aug. 2022
Rosalie Chaleunsouck, Undergraduate Aug. 2021–Aug. 2022

TEACHING EXPERIENCE

Camp Cosmos Instructor, Las Vegas, NV 4 Camps, 2023
Developed and delivered a 5-day summer day camp to Middle School aged students covering topics of astronomy from planets to cosmology.

Lab Instructor , University of Nevada, Las Vegas	2017–2022
Phys. 180L, Mechanics Lab	9 Sections, 5 Terms
Phys. 181L, Electromagnetism Lab	2 Sections, 1 Term
Phys. 151L, General Physics I Lab	6 Sections, 3 Terms
Phys. 152L, General Physics II Lab	5 Sections, 4 Terms
SAGA Education Math Teacher , Chicago, IL	Aug. 2016–Jun. 2017
Delivered a daily, individualized, small-group class in algebra and geometry at Phillips Academy High School with 96% low-income students.	

AWARDS & CERTIFICATIONS

President’s Foundation Graduate Research Fellowship , UNLV	2022
Graduate Mentorship Certificate , UNLV	2022
1st Place, GPSA Research Forum Podium Session , UNLV	2021, 2022
Community Engagement Student Service Honorable Mention , UNLV	2021
Donna Weistrop & David B. Shaffer Scholarship , UNLV	2018, 2019
Nevada Space Grant Graduate Fellowship , NASA	2018
Dean’s List , Northwestern University	Fall 2013, Fall 2014, Spring 2016
Illinois Undergraduate Space Grant , NASA	2015

PROFESSIONAL ACTIVITIES, OUTREACH, AND SERVICE

Conference Organizer

SOC , Layers of Understanding: Model Intercomparisons of Exoplanet Interiors, Max Planck Institute for Astronomy	Apr. 2026
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Workshop/Seminar Organizer

Let’s Build a Planet: Improving Interior Models, Rocky Worlds II Conference, 50 break-out session participants	Jul. 2022
ARCO Seminar Series, OpenU	Sept. 2025–Jan. 2026
Journal Club and Astro Coffee, UNLV	Aug. 2018–May 2019

Journal Referee

ApJ, ApJ Letters, Nat. Astron, Nat. Commun., PSJ, MNRAS, and PASJ.

Journal of Emerging Investigators Referee **Aug. 2024–Present**

Provide constructive and encouraging reviews for high school scientists.

Outreach

Astronomy on Tap, Las Vegas	Jun. 2018–Dec. 2023
Founder, Lead Organizer, and Emcee	
Led quarterly events with over 50 attendees per event	
Skype a Scientist	
Invited video Q&A, 5th grade project	Feb. 2022
Invited video lecture, 5th grade class: “Are Exoplanets Habitable?”	Feb. 2021
Invited video lecture, high school astronomy club	Feb. 2021
Invited video lecture, 11th grade physics class	Apr. 2020

STEM Nova Award Scouts Day Camp Science Advisor and Station Organizer	Jan. 2022
Dallas Center-Grimes High School Invited lecture, AP Physics class	Jan. 2022
Camp Hippocampus Employee Astronomy Night Invited lecture: “Our Cosmic Address”	Sep. 2021
Las Vegas Astronomical Society monthly meeting Invited lecture: “Exoplanets”	Jun. 2020
The CSN Planetarium Astronomy Q&A livestream Invited lecture: “The Pluto Controversy”, [https://youtu.be/EnzSoXZJtIs]	Apr. 2020

Volunteer Service

Judge for American International School Science Fair	May. 2025
Judge for AAS Chambliss Student Achievement Awards	Jan. 2023
Judge for Southern NV Regional Science & Engineering Fair	Mar. 2022
Volunteer Observer with Project RECON	Oct. 2021

PUBLICATIONS

12. **Rice, D.**, Chenliang, H., Royer, R., Daspute, M., Mittal, K. Steffen, J., Vazan, A., (2025) *Magrathea v2: A planetary interior modeling platform in C++*, Submitted to Journal of Open Source Software
11. Steffen, J. H., Shakespeare, C., Royer, R., **Rice, D.**, Vazan, A., (2025) *Effect of Galactic Chemical Evolution on Exoplanet Properties*, **ApJL**, **991**, L33 [[arXiv:2504.16201](https://arxiv.org/abs/2504.16201)].
10. **Rice, D. R.**, Chenliang, H., Steffen, J. H., Vazan, A., (2025) *Uncertainties in the Inference of Internal Structure: The Case of TRAPPIST-1 f*, **ApJ**, **986**, 2 [[arXiv:2504.16201](https://arxiv.org/abs/2504.16201)].
9. **Rice, D. R.**, Steffen, J. H., Vazan, A., (2024) *The Distribution of Planet Radius in Kepler Multiplanet Systems Depends on Gap Complexity*, **ApJL**, **973**, L4 [[arXiv:2406.12239](https://arxiv.org/abs/2406.12239)].
8. Anmol D., Turtelboom, E. V., Harada, C. K., Dressing, C., **Rice, D. R.**, et al., (2024) *The TESS-Keck Survey. XVIII. A Sub-Neptune and Spurious Long-period Signal in the TOI-1751 System*, **AJ**, **167**, 194 [[arXiv:2402.07110](https://arxiv.org/abs/2402.07110)].
7. Childs, A. C., Yang, C., Shakespeare, C., **Rice, D.R.**, Steffen, J. H., (2023) *Composition constraints of the TRAPPIST-1 planets from their formation*, **MNRAS**, **524**, 3748 [[arXiv:2307.04989](https://arxiv.org/abs/2307.04989)].
6. **Rice, D. R.**, Steffen, J. H., (2023) *Stable lifetime of compact, evenly-spaced planetary systems with non-equal masses*, **MNRAS**, **520**, 4057 [[arXiv:2206.11374](https://arxiv.org/abs/2206.11374)].
5. Huang, C. H., **Rice, D. R.**, Steffen, J. H., (2022) *MAGRATHEA: An open-source spherical symmetric planet interior structure code*, **MNRAS**, **513**, 5256 [[arXiv:2201.03094](https://arxiv.org/abs/2201.03094)].
4. MacDonald, M. G., Feil, L., Quinn T., **Rice, D. R.**, (2022) *Confirming the 3:2 Resonant Chain of K2-138*, **AJ**, **163**, 162 [[arXiv:2201.12687](https://arxiv.org/abs/2201.12687)].

3. Huang, C., **Rice, D. R.**, Grande, Z. M., Smith, D., Smith, J. S., Boisvert, J. H., Tschauner, O., Salamat, A., Steffen, J. H., (2021) *Implications of an improved water equation of state for water-rich planets*, **MNRAS**, **503**, **2825** [[arXiv:2103.01410](#)].
2. **Rice, D. R.**, Rasio, F. A., Steffen, J. H., (2018) *Survival of non-coplanar, closely packed planetary systems after a close encounter*, **MNRAS**, **481**, **2205** [[arXiv:1807.07668](#)].
1. DeForest, C. E., Matthaeus, W. H., Howard, T. A., **Rice, D. R.**, (2015) *Turbulence in the solar wind measured with comet tail test particles*, **ApJ**, **812**, **108**

INVITED TALKS

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| 9. ETH Zurich, Exoplanet Journal Club | July. 2025 |
| 8. Indiana University, Exoplanet Group Meeting | Sep. 2024 |
| 7. The Ohio State University, Planet Group Meeting | Sep. 2024 |
| 6. University of Wisconsin, Monday Science Seminar | Sep. 2024 |
| 5. Virtual Astronomy Software Talks, exoVAST | Sep. 2023 |
| 4. The Ohio State University, Exoplanet Talk Series | Oct. 2022 |
| 3. The University of Chicago, Exoplanet Journal Club | Oct. 2022 |
| 2. Northwestern University, Astro Theory Group | Sep. 2022 |
| 1. PennState, The Center for Exoplanets and Habitable Worlds Seminar | Sep. 2022 |

CONTRIBUTED TALKS

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| 8. NASA's Exoplanet Modeling and Analysis Center Workshop, Virtual "Solving and visualizing planet interiors with MAGRATHEA" | Feb. 2023 |
| 7. 241st American Astronomical Society Meeting, Seattle, WA "Interior models of small planets" | Jan. 2023 |
| 6. Exoplanets IV, Las Vegas, NV "Investigating systematic uncertainties in terrestrial interior models with MAGRATHEA", [https://my.aas.org/services/AASTCS9] | May 2022 |
| 5. Graduate & Professional Student Research Forum, UNLV "Systematic uncertainties in terrestrial interior models with MAGRATHEA", [https://youtu.be/6SNhho28NQ0] | Apr. 2022 |
| 4. Graduate & Professional Student Research Forum, UNLV "Characterizing the composition of small exoplanets" | Apr. 2021 |
| 3. Habitable Worlds 2 Conference, Online "MAGRATHEA: Terrestrial planet interior solver and the degeneracy of interiors", [https://youtu.be/1AXe-EvkPcc] | Feb. 2021 |
| 2. Chicago Exoplanet Meeting, The University of Chicago, IL "Dynamical instability in exoplanetary systems" | Jun. 2015 |
| 1. LASP REU, Boulder, CO "Analysis of comet tails for turbulence in the solar wind", | Jul. 2014 |

CONTRIBUTED POSTERS

7. Rocky Worlds 4, Groningen, Netherlands **Jan. 2026**
“From Single Planet Inference to Population Trends in Rocky Exoplanet Interiors”
6. Extreme Solar Systems V, Christchurch, NZ **Mar. 2024**
“MAGRATHEA 2.0: Advancements in Interior Structure Modeling for Exoplanets”
5. Rocky Worlds II, Oxford, UK **Jul. 2022**
“Investigating systematic uncertainties in terrestrial interior models with MAGRATHEA”
4. Exoplanets III, Online **Jul. 2020**
“MAGRATHEA: Terrestrial planet interior solver and the degeneracy of interiors”,
[<https://www.physics.unlv.edu/~drice986/Exo3Poster/riceexoposter.html>]
3. Sagan Exoplanet Summer Workshop, Pasadena, CA **Jul. 2019**
“Differentiated collisions and their effect on terrestrial planet composition”
2. Kepler & K2 Science Convention V, Glendale, CA **Mar. 2019**
“The effect of differentiated collisions on the interiors of terrestrial planets”
1. 227th American Astronomical Society Meeting, Kissimmee, FL **Jan. 2016**
“Understanding dynamical instability in 4-planet systems with equal orbital spacing”